

Adding Vectors: Graphical and Component Methods

PhET Vector Addition · PhET

Senior Secondary (Years 11-12)

~30 minutes

Mathematics · Specialist
Mathematics

NOTICE — AI-generated worksheet

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Simulation: <https://phet.colorado.edu/en/simulations/vector-addition>

Curriculum links: proof · vectors

Learning intentions

- I can add two or three vectors graphically using the tip-to-tail method.
- I can calculate a resultant vector's components and magnitude using trigonometry.
- I can verify a graphical vector sum against a calculated result.

Getting started

Open Vector Addition and start on the Explore 2D screen. Drag two vectors onto the grid, join them tip-to-tail, and note where the sum (resultant) vector ends up, before turning on the component grid to check your answer numerically.

Tasks

1. Place two vectors on the grid at different angles. Estimate the resultant's direction and length just by looking at the tip-to-tail diagram.

2. Turn on the component display. Record the x- and y-components of each of your two vectors, then add the components to calculate the resultant vector's components.

Working:

3. Use Pythagoras' theorem and trigonometry to calculate the magnitude and direction (angle) of your resultant vector from its components.

Working:

4. Compare your calculated resultant to the one shown on the grid. How closely do they match?

5. Repeat the addition for three different pairs of vectors and record each pair's calculated resultant magnitude and angle.

Vector pair	Resultant magnitude	Resultant angle

6. Explain why adding vectors tip-to-tail on a diagram gives the same result as adding their components algebraically.

7. Add three vectors instead of two. Show your component working and compare the calculated resultant to the diagram.

Working:

8. Explain what would need to be true about three vectors for their resultant to equal zero, and describe how you could check this using the simulation.

Extension (optional)

A plane flies with a velocity vector relative to the air, and the wind has its own velocity vector. Use vector addition to explain how you would find the plane's actual velocity relative to the ground, and what this means for the difference between the pilot's heading and their actual direction of travel.
